Burns-Epstein invariant and its gradient flow

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Overview

1. Cauchy-Riemann geometry
   - CR manifolds
   - Local invariants for pseudo-Hermtian structure
   - Local invariants for CR structure
2. Burns-Epstein invariant
   - Chern-Simons 3-form and global CR invariant
   - Variational aspect
3. Gradient flow
   - Negative gradient flow equation
   - Short-time existence

Cauchy-Riemann manifolds

Definition

A CR manifold M is a real $(2n+k)$-dimensional smooth manifold equipped with a CR structure (ξ, J) where $\xi \subset TM$ is a rank-$2n$ subbundle and $J \in C^\infty(M, \mathrm{End}(\xi))$ is a complex structure assigned to ξ such that for any $X, Y \in C^\infty(M, \xi)$ the following conditions hold

i) $[X, Y] - [JX, JY] \in C^\infty(M, \xi)$

ii) $\mathcal{N}_J(X, Y) = J([X, Y] - [JX, JY]) - ([JX, Y] + [X, JY]) = 0$

We say (M, ξ, J) is a CR codimension k manifold.

Note: ξ in general can be either integrable or non-integrable.

Cauchy-Riemann manifolds

Alternative definition of CR manifold: $(M^{2n+k}, \xi^{1,0})$ with complex rank-n subbundle $\xi^{1,0} \subset T_{\mathbb{C}}M$ called the *holomorphic tangent bundle* satisfying:
i) $\xi^{1,0} \cap \xi^{0,1} = \{0\}$, $\xi^{1,0} \oplus \xi^{0,1} = \xi_{\mathbb{C}}$, $\mathrm{Re}(\xi^{1,0} \oplus \xi^{0,1}) = \xi$.
ii) $[\xi^{1,0}, \xi^{1,0}] \subset \xi^{1,0}$.

Recovering to the original definition:

$$\xi^{1,0}_x = \{Z = X - \mathrm{i}JX : X \in \xi_x\} = \ker(J - \mathbf{i}\mathrm{Id})$$
$$\xi^{0,1}_x = \overline{\xi^{1,0}_x} = \left\{\bar{Z} = X + \mathrm{i}JX : X \in \xi_x\right\} = \ker(J + \mathbf{i}\mathrm{Id})$$

Cauchy-Riemann manifolds

In the CR codimension 1 case, when M is orientable, there exists a global 1-form θ s.t. $\xi = \ker(\theta)$. Such θ is referred as a *pseudo-Hermitian structure*. The underlying hyperplane distribution ξ is now actually a *contact structure*, its defining form θ is also called a contact form. The splitting of complexified tangent bundle: $T_{\mathbb{C}}M = \xi^{1,0} \oplus \xi^{0,1} \oplus \mathbb{C}\mathcal{R}$ where $\mathcal{R}$ is called the Reeb vector field.

There is an induced Hermitian form on $\xi_{\mathbb{C}}$:

$$L_\theta(Z, \bar{Z}') := -\mathrm{i}d\theta(Z, \bar{Z}') \qquad \forall\, Z \in \xi^{1,0}, \bar{Z}' \in \xi^{0,1} \tag{1}$$

This is called the *Levi form*. It is the $\mathbb{C}$-linear extension of bilinear form $d\theta(\cdot, J\cdot)$ on ξ to $\xi_{\mathbb{C}}$. If $\tilde{\theta} = f\theta$, then $L_{\tilde{\theta}} = fL_\theta$.

Definition

If the Levi form is non-degenerate, the CR structure is called non-degenerate. If the Levi form is further positive or negative definite, then the CR structure is called strictly pseudo-convex or strictly J-convex.

Cauchy-Riemann manifolds

Definition

• A smooth map $F : M_1 \to M_2$ is a CR map with respect to CR structures $(M_1, \xi_1^{1,0})$, $(M_2, \xi_2^{1,0})$ if $dF(\xi_1^{1,0}) \subset \xi_2^{1,0}$.

• If for any point $p \in M_1$, there is a neighborhood $U_p \subseteq M_1$ and on U_x a diffeomorphism $F : U_p \subseteq M_1 \to V_y \subseteq M_2$, $y = F(x)$ satisfies $dF(\xi_1^{1,0}|_{U_x}) = \xi_2^{1,0}|_{V_y}$, then $(M_1, \xi_1^{1,0})$, $(M_2, \xi_2^{1,0})$ are CR equivalent. If such diffeomorphism exists globally, then it is called CR diffeomorphism.

Alternative description: A smooth map F between CR manifolds (M_1, ξ_1, J_1), (M_2, ξ_2, J_2) is a CR map if and only if $dF(\xi_1|_x) \subset \xi_2|_{F(x)}$ and $dF \circ J_1 = J_2 \circ dF$ on ξ_x for every $x \in M$.

Fact: The (local) CR diffeomorphisms between real analytic CR hypersurfaces in $\mathbb{C}^{n+1}$ can be typically given by (local) biholomorphisms.

Local invariants for pseudo-Hermtian structure

Let (M, ξ, J) be a strictly pseudo-convex CR manifold of $2n+1$-dimensional, and θ is a fixed pseudo-Hermitian structure on it. One may extend J to a bundle endomorphism $J : TM \to TM$ by putting $J\mathcal{R} = 0$. In this case, there is an induced Riemannian metric called *Webster metric* such that

$$g_\theta(X, Y) = d\theta(X, JY)\,,\ g_\theta(X, \mathcal{R}) = 0\,,\ g_\theta(\mathcal{R}, \mathcal{R}) = 1 \qquad \forall\, X, Y \in \xi.$$

The CR manifold becomes a Riemannian manifold. The data $(J, \mathcal{R}, \theta, g_\theta)$ is called a *contact metric structure* on (M, ξ).

Local invariants for pseudo-Hermtian structure

Theorem

Let $(M, \xi^{1,0})$ be a strictly pseudo-convex CR manifold and θ is a fixed pseudo-Hermitian structure. J is a complex structure extended to TM as above, and g_θ is the corresponding Webster metric. There is a unique affine connection ∇ on M such that:
i) ξ is parallel with respect to ∇, i.e. $\nabla_X C^\infty(M, \xi) \subseteq C^\infty(M, \xi)$ $\forall\, X \in TM$.
ii) $\nabla J = 0$ and $\nabla g_\theta = 0$ (namely J and g_θ are also ∇-parallel).
iii) T^∇ is pure.

The torsion is said to be *pure* if it satisfies the conditions: For all $Z, Z' \in \xi^{1,0}$,

$$T^\nabla(Z, Z') = 0\,, \quad T^\nabla(Z, \bar{Z}') = 2d\theta(Z, \bar{Z}')\mathcal{R}\,, \tau \circ J + J \circ \tau = 0$$

where $\tau : TM \to TM$ is defined by $\tau(X) = T^\nabla(\mathcal{R}, X)$ for any $X \in TM$.

Local invariants for pseudo-Hermitian structure

Webster 1978: Pseudo-Hermitian structure as a integrable $\mathrm{U}(p,q)$-structure

Cartan moving frame method: Pick local frame field $\{Z_1,\dots,Z_n\}$ of $\xi^{1,0}$, $\nabla Z_\beta = \omega_\beta^\alpha \otimes Z_\alpha$, $\tau Z_\beta = A_\beta^{\bar\alpha} Z_{\bar\alpha}$.
$\Rightarrow$ Coframe $\{\theta^\alpha, \theta^{\bar\alpha}, \theta\}$: $\theta^\alpha(Z_\beta) = \delta_\beta^\alpha, \theta^\alpha(Z_{\bar\beta}) = \theta^\alpha(\mathcal{R}) = 0$.

Pseudo-Hermitian structure equation: $d\theta = \mathrm{i} h_{\alpha\bar\beta}\theta^\alpha \wedge \theta^{\bar\beta}$

Cartan's first structure equation: $d\theta^\alpha = \theta^\beta \wedge \omega_\beta^\alpha + \theta \wedge \tau^\alpha$

Cartan's second stricture equation: $\Omega_\alpha^\beta = d\omega_\alpha^\beta - \omega_\alpha^\gamma \wedge \omega_\gamma^\beta$.

Theorem

The curvature 2-form is expressed as

$$\Omega_\alpha^\beta = R_\alpha{}^\beta{}_{\rho\bar\sigma}\theta^\rho \wedge \theta^{\bar\sigma} + W_{\alpha\rho}^\beta \theta^\rho \wedge \theta - W_{\alpha\bar\sigma}^\beta \theta^{\bar\sigma} \wedge \theta + \mathrm{i}\theta_\alpha \wedge \tau^\beta - \mathrm{i}\tau_\alpha \wedge \theta^\beta,$$

where $R_\alpha{}^\beta{}_{\rho\bar\sigma}$ is called the pseudo-Hermitian curvature tensor and $W_{\alpha\bar\sigma}^\beta = h^{\bar\gamma\beta} A_{\bar\sigma\bar\gamma,\alpha}$, $W_{\alpha\rho}^\beta = h^{\bar\gamma\beta} A_{\beta\rho,\bar\gamma}$ the Webster curvature tensor.

Local invariants for pseudo-Hermitian structure

3-dimensional case: coframes $\{\theta^1, \theta^{\bar{1}}\}$ form U(1)-structure $\mathcal{X} \to M$

Pseudo-Hermitian structure equation:

$$d\theta = \mathrm{i}\theta^1 \wedge \theta^{\bar{1}}$$

Connection and torsion:

$$d\theta^1 = \theta^1 \wedge \theta^1_1 + \theta \wedge \tau^1$$

with normalization $\theta^1_1 + \theta^{\bar{1}}_{\bar{1}} = 0$ and $\tau^1 \wedge \theta^{\bar{1}} = 0$.

Curvature:

$$d\theta^1_1 = R\theta^1 \wedge \theta^{\bar{1}} + W\theta^1 \wedge \theta - \overline{W}\theta^{\bar{1}} \wedge \theta$$

$$W = \nabla_{Z_1} A^{\bar{1}}_1 \quad , \quad \overline{W} = \nabla_{Z_{\bar{1}}} A^1_{\bar{1}}.$$

Local invariants for CR structure

Chern-Moser 1974: CR structure as an integrable G-structure

In 3-dimensional case: Consider positive part of full symplectization

$$S_+(M,\xi) := \{(x, u\theta_x) : x \in M\,,\ \ker(\theta_x) = \xi_x\,,\ u \in \mathbb{R}_+\} \subset T^*M$$

coframes $\{\Theta = u\theta, \Theta^1 = \sqrt{u}\theta^1, \Theta^{\bar{1}} = \sqrt{u}\theta^{\bar{1}}, \phi = -\frac{du}{u} + \varphi\}$ on $S_+(M,\xi)$ form a G-structure $\mathcal{Y} \to S_+(M,\xi)$ with CR structure equation

$$d\Theta = \mathrm{i}\Theta^1 \wedge \Theta^{\bar{1}} + \Theta \wedge \phi$$

$G = G_1 \subset \mathbf{SU}(2,1) \subset \mathrm{GL}(4,\mathbb{C})$, $\mathfrak{su}(2,1) = \mathfrak{g} \oplus \mathfrak{m}$.

Theorem

There is a uniquely defined torsion-free $\mathfrak{su}(2,1)$-valued Cartan connection $\omega \in \Omega^1(\mathcal{Y}; \mathfrak{su}(2,1))$ on $\mathcal{Y}$ with $\mathrm{tr}\,\Omega = 0$.

Torsion-free here means: $\mathrm{pr}_{\mathfrak{m}}\,\Omega = 0$.

Loacl invariants for CR structure

Along local section (moving coframe) $\sigma : U \subset S_+(M, \xi) \to \mathcal{Y}$
$\pi = \sigma^*\omega$, $\Pi = \sigma^*\Omega$

$$\pi = \begin{pmatrix} \pi_0^0 & \pi_0^1 & \pi_0^2 \\ \pi_1^0 & \pi_1^1 & \pi_1^2 \\ \pi_2^0 & \pi_2^1 & \pi_2^2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{3}(\phi_1^1 + \phi) & \Theta^1 & 2\Theta \\ -\mathrm{i}\phi^{\bar{1}} & \frac{1}{3}(\phi_1^1 - \phi_{\bar{1}}^{\bar{1}}) & 2\mathrm{i}\Theta^{\bar{1}} \\ -\frac{1}{4}\psi & \frac{1}{2}\phi^1 & \frac{1}{3}(\phi + \phi_{\bar{1}}^{\bar{1}}) \end{pmatrix}$$

$$\Pi = d\pi + \pi \wedge \pi$$

$$= \begin{pmatrix} \Pi_0^0 & 0 & 0 \\ \Pi_1^0 & \Pi_1^1 & 0 \\ \Pi_2^0 & \Pi_2^1 & \Pi_2^2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ -\mathrm{i}Q_1^{\bar{1}}\Theta^1 \wedge \Theta & 0 & 0 \\ -\frac{1}{4}(R_1\Theta^1 + R_{\bar{1}}\Theta^{\bar{1}}) \wedge \Theta & \frac{1}{2}Q_{\bar{1}}^1\Theta^{\bar{1}} \wedge \Theta & 0 \end{pmatrix}$$

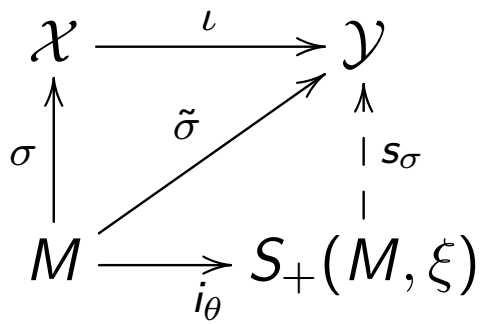
Relating local pseudo-Hermitian invariants to local CR invariants

$$\{(\theta^1, \theta^{\bar{1}}) : d\theta = \mathrm{i}\theta^1 \wedge \theta^{\bar{1}}\} \overset{1:1}{\leftrightarrow} \{(\Theta, \Theta^1, \Theta^{\bar{1}}, \phi) : \Theta = \mathrm{pr}^*\theta, u = 1, \phi = 0\}$$

Proposition

There is a bundle inclusion $\iota : \mathcal{X} \to \mathcal{Y}$ with $\iota|_M = \mathrm{id}_M$. In particular, it is a homotopy equivalence.

$\Rightarrow$



Relating local pseudo-Hermitian invariants to local CR invariants

$\Rightarrow$

$$\phi_1^1 = \theta_1^1 + \frac{\mathrm{i}R}{4}\theta$$

$$\phi^1 = \frac{\mathrm{i}R}{4}\theta^1 + E\theta + \tau^1$$

$$\psi = G\theta + \mathrm{i}(\bar{E}\theta^1 - E\theta^{\bar{1}})$$

$$E = \frac{1}{6}R_{,1} - \frac{2\mathrm{i}}{3}A^{\bar{1}}_{1,\bar{1}}$$

$$Q_1^{\bar{1}} = \frac{1}{6}R_{,1}{}^{\bar{1}} + \frac{\mathrm{i}}{2}RA^{\bar{1}}_{\bar{1}} - A^{\bar{1}}_{1,0} - \frac{2\mathrm{i}}{3}A^{\bar{1}}_{1,\bar{1}}{}^{\bar{1}}.$$

Chern-Simons 3-form and global CR invariant

$C_2(\Pi) = \frac{1}{8\pi^2}[\mathrm{tr}(\Pi \wedge \Pi) + \mathrm{tr}(\Pi) \wedge \mathrm{tr}(\Pi)] = \frac{1}{8\pi^2}\mathrm{tr}(\Pi \wedge \Pi) \in H^4(S_+(M,\xi);\mathbb{R})$
$C_2(\Pi) \equiv 0 \Rightarrow dTC_2(\Pi) = C_2(\Pi)$ ("transgression")
Chern-Simons 3-form:

$$TC_2(\pi) = \frac{1}{8\pi^2}\left[\mathrm{tr}(\pi \wedge \Pi) + \frac{1}{3}\mathrm{tr}(\pi \wedge \pi \wedge \pi)\right] = s^* TC_2(\omega) \in H^3(S_+(M,\xi);\mathbb{R}),$$

Proposition

$[TC_2(\pi)] = s^*[TC_2(\omega)]$ is independent of the choice of section $s : S_+(M,\xi) \to \mathcal{Y}$.

Proof. Let s and s' be arbitrary two sections so that $s' = s \cdot h$ pointwise with some $h \in G_1$.
Computation shows $s^* \mathrm{tr}(\omega \wedge \Omega) = \mathrm{tr}(\pi \wedge \Pi) = 0$
$\Rightarrow TC_2(\omega) = \frac{1}{8\pi^2}\left[\mathrm{tr}(\omega \wedge \Omega) + \frac{1}{3}\mathrm{tr}(\omega \wedge \omega \wedge \omega)\right] = \frac{1}{24\pi^2}\mathrm{tr}(\omega \wedge \omega \wedge \omega)$

Chern-Simons 3-form and global CR invariant

Put $\pi' = s'^*\omega$, by the connection transformation formula $\pi' = \mathrm{Ad}(h^{-1})\pi + h^{-1}dh$ and the trace property for product of matrices, we can derive out of computation that

$$TC_2(\pi') - TC_2(\pi) = -\frac{1}{8\pi^2} d\,\mathrm{tr}(h^{-1}\pi \wedge dh).$$

This is an exact form. Hence $[TC_2(\pi)] = s^*[TC_2(\omega)]$ is independent of choice of section s. □

Remark

$[TC_2(\pi)]$ is a CR invariant class on $S_+(M, \xi)$.

Chern-Simons 3-form and global CR invariant

Goal: pull down $[TC_2(\pi)]$ to a CR invariant class on M that can be expressed in terms of local pseudo-Hermitian invariants.

- $i_\theta : M \hookrightarrow S_+(M,\xi) \cong M \times \mathbb{R}_+$ is a homotopy equivalence, thus $H^3(S_+(M,\xi);\mathbb{R}) \cong H^3(M;\mathbb{R})$. $i_\theta^*[TC_2(\pi)] \in H^3(M;\mathbb{R})$. This class is independent of choice of contact form θ: $i_\theta^*[TC_2(\pi)] = i_{\theta'}^*[TC_2(\pi)] \in H^3(M;\mathbb{R})$.

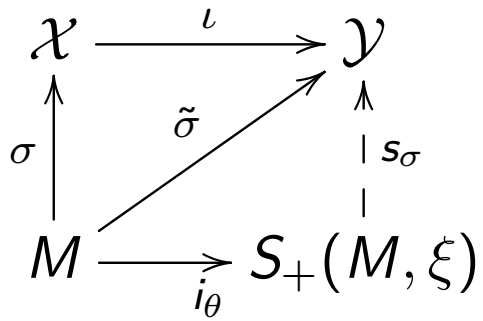
- The bundle inclusion $\iota : \mathcal{X} \hookrightarrow \mathcal{Y}$ is a homotopy equivalence $\Rightarrow$ $H^3(\mathcal{Y};\mathbb{R}) \cong H^3(\mathcal{X};\mathbb{R}) \Rightarrow$ CR invariant class $\iota^*[TC_2(\omega)] \in H^3(\mathcal{X};\mathbb{R})$ on $\mathcal{X}$. From Gysin sequence for $\mathcal{X} \xrightarrow{p} M$, we have S.E.S.

$$0 \to H^3(M;\mathbb{R}) \xrightarrow{p^*} H^3(\mathcal{X};\mathbb{R}) \xrightarrow{p_*} H^2(M;\mathbb{R}) \to 0$$

$\sigma^* : H^3(\mathcal{X};\mathbb{R}) \to H^3(M;\mathbb{R})$ is a left inverse of $p^* \Rightarrow$ S.E.S. splits, i.e. $H^3(\mathcal{X};\mathbb{R}) \cong H^3(M;\mathbb{R}) \oplus H^2(M;\mathbb{R})$. Pulling $\iota^*[TC_2(\omega)]$ back to M along σ means dropping the part of $H^2(M;\mathbb{R})$ along with a circle factor from p_* evaluating on the class in $H^1(\mathbb{S}^1;\mathbb{R}) \cong \mathbb{R}$.

Chern-Simons 3-form and global CR invariant

• $[\tilde{T}C_2(\pi)] := \sigma^*\iota^*[TC_2(\omega)]$ is independent of the choice of σ:
By the commutative diagram



$\Rightarrow [\tilde{T}C_2(\pi)] = i_\theta^* s_\sigma^* [TC_2(\omega)] = i_\theta^*[TC_2(\pi)]$ where
$\pi = s_\sigma^*\omega \in \Omega^1(S_+(M,\xi);\mathfrak{su}(2,1))$. $\sigma, \sigma' : M \to \mathcal{X}$ induce
$s_\sigma, s_{\sigma'} : S_+(M,\xi) \to \mathcal{Y}$, previous Proposition implies
$s_\sigma^*[TC_2(\omega)] = s_{\sigma'}^*[TC_2(\omega)] \in H^3(S_+(M,\xi);\mathbb{R})$.

In conclusion, $[\tilde{T}C_2(\pi)]$ is indeed a CR invariant class on M.

$$[\tilde{T}C_2(\pi)] = \left[\frac{\mathrm{i}}{8\pi^2}\left(-\frac{2\mathrm{i}}{3}d\theta_1^1 \wedge \theta_1^1 + \frac{1}{6}R\,\theta \wedge d\theta_1^1 - 2\theta \wedge \tau^1 \wedge \tau^{\bar{1}}\right)\right] \in H^3(M;\mathbb{R}).$$

Chern-Simons 3-form and global CR invariant

Definition (Burns-Epstein 1988)

On oriented closed CR 3-manifold (M, ξ, J) with trivial holomorphic tangent bundle, the following quantity is globally well-defined CR invariant:

$$\mu(J) = \int_M \tilde{T} C_2(\pi) = \frac{1}{8\pi^2} \left[\int_M \left(\frac{1}{6} R^2 + 2|A_{\bar{1}}^1|^2 \right) \theta \wedge d\theta + \int_M \frac{2}{3} \theta_1^1 \wedge d\theta_1^1 \right]$$

Definition (Cheng-Lee 1990)

On oriented closed contact 3-manifold (M, ξ) with any two $J, \hat{J}$, the following quantity is globally well-defined and independent of choice of θ:

$$\mu(J, \hat{J}) = \frac{1}{8\pi^2} \left[\int_M \left(\frac{1}{6} R^2 + 2|A|^2 \right) \theta \wedge d\theta - \int_M \left(\frac{1}{6} \hat{R}^2 + 2|\hat{A}|^2 \right) \theta \wedge d\theta + \int_M \frac{2}{3} (\theta_1^1 - \hat{\theta}_1^1) \wedge (d\theta_1^1 + d\hat{\theta}_1^1) \right]$$

Remark

When holomorphic tangent bundles are trivial, $\mu(J, \hat{J}) = \mu(J) - \mu(\hat{J})$.

Variational aspect

Let $\mathcal{J} \subset \mathrm{End}(\xi)$ be the bundle of oriented CR structures.

$\mu(J, \hat{J})$ with fixed $\hat{J}$ is a functional $C^\infty(M, \mathcal{J}) \to \mathbb{R}$.

For any smooth section $J \in C^\infty(M, \mathcal{J})$, define a subbundle

$$\mathcal{E}_J := \{E \in \mathrm{End}(\xi) : E \circ J + J \circ E = 0\} \subset \mathrm{End}(\xi)$$

Deformation of CR structure:

$$\{J_t \in C^\infty(M, \mathcal{J}) : -\varepsilon < t < \varepsilon\} \subset C^\infty(M, \mathcal{J})$$

Tangent space:

$$T_J C^\infty(M, \mathcal{J}) = \{E \in C^\infty(M, \mathrm{End}(\xi)) \,|\, E = \dot{J}\} = C^\infty(M, \mathcal{E}_J)$$

where $\dot{J} = \frac{d}{dt}\big|_{t=0} J_t$, $J_t \in C^\infty(M, \mathcal{J})\ \forall\, t \in (-\varepsilon, \varepsilon)$.

Variational aspect

L^2-inner product on $C^\infty(M, \mathcal{J})$ via the volume integral

$$(\cdot,\cdot)_J = \int_M \langle \cdot,\cdot \rangle_J \theta \wedge d\theta$$

with Levi form $\langle \cdot,\cdot \rangle_J = d\theta(\cdot, J\cdot)_{\mathbb{C}}$ s.t. $\langle V, W \rangle_J = h_{1\bar{1}} V^1 W^{\bar{1}}$ for $V = V^1 Z_1$, $W = W^1 Z_1$ in $\xi^{1,0}$.

Proposition

The first order variation of Burns-Epstein functional with respect to deformation J_t of J and $\hat{J}$ fixed is given by

$$D\mu(J,\hat{J})|_J = \frac{d}{dt}\Big|_{t=0} \mu(J_t,\hat{J}) = -\frac{1}{8\pi^2}(\dot{J}, Q_J)_J.$$

where $Q_J = \mathrm{i}Q_1^{\bar{1}}\theta^1 \otimes Z_{\bar{1}} - \mathrm{i}Q_{\bar{1}}^1 \theta^{\bar{1}} \otimes Z_1$.

Q_J is called the Cartan tensor.

Variational aspect

Proof. Fix a unit local frame Z_1 in $\xi^{1,0}$ with dual coframe θ^1. Let $\dot{J} = 2\dot{E} = 2\dot{E}_1^{\bar{1}}\theta^1 \otimes Z_{\bar{1}} + 2\dot{E}_{\bar{1}}^{1}\theta^{\bar{1}} \otimes Z_1$. $1 = h_{1\bar{1}\,t} = 2 - \mathrm{i}d\theta(Z_{1\,t}, Z_{\bar{1}\,t})$.

$$\Rightarrow 0 = -2\mathrm{i}[d\theta(\dot{Z}_1, Z_{\bar{1}}) + d\theta(Z_1, \dot{Z}_{\bar{1}})] \Rightarrow \dot{\theta}^1 = -\mathrm{i}\dot{E}_{\bar{1}}^{1}\theta^{\bar{1}}$$

$$d\dot{\theta}^1 = \dot{\theta}^1 \wedge \theta_1^1 + \theta^1 \wedge \dot{\theta}_1^1 + \dot{A}_{\bar{1}}^{1}\theta \wedge \theta^{\bar{1}} + A_{\bar{1}}^{1}\theta \wedge \dot{\theta}^{\bar{1}} \Rightarrow \dot{A}_{\bar{1}}^{1} = -\mathrm{i}\dot{E}_{\bar{1},0}^{1}$$

$$\Rightarrow \dot{\theta}_1^1 = \mathrm{i}(A_1^{\bar{1}}\dot{E}_{\bar{1}}^{1} + A_{\bar{1}}^{1}\dot{E}_1^{\bar{1}})\theta - \mathrm{i}(\dot{E}_{1,1}^{\bar{1}}\theta^1 + \dot{E}_{\bar{1},1}^{1}\theta^{\bar{1}})$$
$$\dot{R} = \mathrm{i}(\dot{E}_{1,\bar{1}}^{\bar{1}}{}^{1} - \dot{E}_{\bar{1},1}^{1}{}^{\bar{1}}) - (A_1^{\bar{1}}\dot{E}_{\bar{1}}^{1} + A_{\bar{1}}^{1}\dot{E}_1^{\bar{1}}).$$

$$\begin{aligned}
\Rightarrow 8\pi^2 \frac{d}{dt}\Big|_{t=0} \mu(J_t, \hat{J}) &= \int_M \left(\frac{1}{3}R\dot{R} + 2\dot{A}_1^{\bar{1}}A_{\bar{1}}^{1} + 2\dot{A}_{\bar{1}}^{1}A_1^{\bar{1}}\right)\theta \wedge d\theta + \frac{2}{3}\int_M \dot{\theta}_1^1 \wedge (d\theta_1^1 + d\hat{\theta}_1^1) + \frac{2}{3}\int_M (\theta_1^1 - \hat{\theta}_1^1) \wedge d\dot{\theta}_1^1 \\
&= 2\,\mathrm{Re}\int_M \left(\frac{1}{6}R\dot{R} + 2A_1^{\bar{1}}\dot{A}_{\bar{1}}^{1}\right)\theta \wedge d\theta + \frac{4}{3}\int_M \dot{\theta}_1^1 \wedge d\theta_1^1 \\
&= 4\,\mathrm{Re}\int_M \mathrm{i}\dot{E}_{\bar{1}}^{1}\left(-\frac{1}{6}R_{,1}{}^{\bar{1}} - \frac{\mathrm{i}}{2}RA_1^{\bar{1}} + A_{1,0}^{\bar{1}} + \frac{2\mathrm{i}}{3}A_{1,1}^{\bar{1}}{}^{1}\right)\theta \wedge d\theta \\
&= -(2\dot{E}, Q_J)_J = -(\dot{J}, Q_J)_J
\end{aligned}$$

Variational aspect

Theorem (Cartan, Chern-Moser)

Let (M, θ) be a strictly pseudo-convex pseudo-Hermitian manifold of dimension $2n+1$.
- If $2n+1=3$, then M is locally CR spherical if and only if the Cartan tensor

$$Q_{\alpha\beta} = \mathrm{i}(A_{\alpha\beta,0} - 2\phi_{\alpha,\beta}) + 2P_{\alpha\bar{\nu}}A^{\bar{\nu}}_{\beta}$$

vanishes, where

$$\phi_\alpha = \frac{1}{n+2}\left(\frac{R_\alpha}{2n+2} - \mathrm{i}A_{\alpha\beta,\bar{\nu}}h^{\beta\bar{\nu}}\right)\ ,\ P_{\alpha\bar{\beta}} = \frac{1}{n+2}\left(R_{\alpha\bar{\beta}} - \frac{R}{2n+2}h_{\alpha\bar{\beta}}\right)\ ,\ R = h^{\alpha\bar{\beta}}R_{\alpha\bar{\beta}}\ ,\ R_{\alpha\bar{\beta}} = -R_{\alpha}{}^{\sigma}{}_{\bar{\beta}\sigma}.$$

- If $2n+1 \geq 5$, then M is locally CR spherical if and only if the Chern tensor below vanishes:

$$C_{\mu}{}^{\beta}{}_{\bar{\nu}\alpha} = -R_{\mu}{}^{\beta}{}_{\bar{\nu}\alpha} - \frac{1}{m+2}(R^{\beta}_{\alpha}h_{\mu\bar{\nu}} + R^{\beta}_{\mu}h_{\alpha\bar{\nu}} + \delta^{\beta}_{\alpha}R_{\mu\bar{\nu}} + \delta^{\beta}_{\mu}R_{\alpha\bar{\nu}}) + \frac{R}{(m+1)(m+2)}(\delta^{\beta}_{\alpha}h_{\mu\bar{\nu}} + \delta^{\beta}_{\mu}h_{\alpha\bar{\nu}}).$$

$\Rightarrow$ The critical point of Burns-Epstein invariant is the spherical CR structure on (M, ξ).

Negative gradient flow equation

The negative L^2-gradient flow equation:

$$\partial_t J_t = -16\pi^2 \operatorname{grad} \mu(J_t, \hat{J}) = 2Q_{J_t}$$

The negative gradient flow is therefore called the Cartan flow.

μ is strictly decreasing in time along the Cartan flow (unless at equilibrium $Q_{J_0} = 0$):

$$\frac{d}{dt}\Big|_{t=0} \mu(J_t, \hat{J}) = -\frac{1}{4\pi^2} \| Q_{J_t} \|^2_{J_t} \Big|_{t=0}$$

2nd order differential operator $B_J : C^\infty(M, \mathrm{End}(\xi)) \to C^\infty(M)$ defined locally by

$$B_J E = E^{\bar{1}}_{1,\bar{1}}{}^{1} + E^{1}_{\bar{1},1}{}^{\bar{1}} - \mathrm{i} A^1_{\bar{1}} E^{\bar{1}}_1 + \mathrm{i} A^{\bar{1}}_1 E^1_{\bar{1}}$$

Formal adjoint $B_J^* : C^\infty(M) \to C^\infty(M, \mathrm{End}(\xi))$ given locally by

$$B_J^* f = (f_{,1}{}^{\bar{1}} + \mathrm{i} A^{\bar{1}}_1 f)\theta^1 \otimes Z_{\bar{1}} + (f_{,\bar{1}}{}^{1} - \mathrm{i} A^1_{\bar{1}} f)\theta^{\bar{1}} \otimes Z_1$$

Negative gradient flow equation

Lemma

Let $f \in C^2(M)$, $J \in C^\infty(M, \mathcal{J})$, and X_f be the corresponding contact vector field. Then $L_{X_f} J = 2B_J^* f$.

Remark

X_f (infinitesimal contactomorphism) is an infinitesimal CR diffeomorphism if and only if $B_J^* f = 0$.

Fact (not easy): $C^\infty(M, \mathcal{E}_{\hat{J}}) = \mathrm{im}(B_{\hat{J}}^*) \oplus \ker(B_{\hat{J}})$

Second variation of $\mu(J, \hat{J})$ needs to be computed w.r.t. to the direction in $\ker(B_{\hat{J}})$.

Negative gradient flow equation

$\partial_t J_t = 2Q_{J_t}$ is a non-linear equation on J_t.

For any contactomorphism φ s.t. $\varphi^*\theta = e^{2f}\theta$ for some $f \in C^\infty(M)$, we have $Q_{\varphi^* J_t} = e^{4f}\varphi^* Q_{J_t}$ (Q_{J_t} is pseudo-Hermitian invariant tensor).

$\Rightarrow$ very large kernel of the principal symbol of the linearized operator DQ_J which acts on $E \in C^\infty(M, \mathcal{E}_J)$ as

$$2\,\mathrm{Re}\left[\left(-\frac{1}{6}E^{\bar{1}}_{1,\bar{1}}{}^{1}{}_{1}{}^{\bar{1}} + \frac{1}{6}E^{1}_{\bar{1},1}{}^{\bar{1}}{}_{1}{}^{\bar{1}} + E^{\bar{1}}_{1,00} + \frac{2}{3}\mathrm{i}E^{\bar{1}}_{1,0}{}^{1}{}_{1}\right)\theta^1 \otimes Z_{\bar{1}} + \mathcal{O}_3(E)\right]$$

i.e. highly degenerate nonlinear PDE.

Negative gradient flow equation

Cheng-Lee 1990: Applied DeTurck's trick regularizing ("gauge fixing") the Cartan flow equation.

Define a quadratic operator $F_J : C^\infty(M, \mathrm{End}(\xi)) \to C^\infty(M)$ given locally by

$$F_J E = \mathrm{i} E_1^1 E_{1,\bar{1}}^{\bar{1}}{}^1 - \mathrm{i} E_{\bar{1}}^{\bar{1}} E_{\bar{1},1}^1{}^{\bar{1}} + \mathrm{i} E_{\bar{1}}^1 E_{1,\bar{1}}^{\bar{1}}{}^{\bar{1}} - \mathrm{i} E_1^{\bar{1}} E_{\bar{1},1}^1{}^1$$

pick any $K \in C^\infty(M, \mathcal{J})$ as a fixed background CR structure $\Rightarrow$
$\mathcal{P} : C^\infty(M, \mathcal{J}) \to C^\infty(M, \mathrm{End}(\xi))$ given by

$$\mathcal{P}(J) = -2Q_J + \frac{1}{6} B_J^* F_J K.$$

It turns out to be a 4th order non-linear subelliptic operator.

Regularized Cartan flow equation:

$$\partial_t J_t + \mathcal{P}(J_t) = 0$$

Negative gradient flow equation

Recover the (unregularized) Cartan flow from the regularized flow in a renormlization way:

Proposition

Suppose J_t is a C^k-solution with $k \geq 8$ to the regularized Cartan flow equation on $[0, \varepsilon]$. Then there exists a C^{k-4}-solution $\tilde{J}_t$ on $[0, \varepsilon]$ to $\partial \tilde{J}_t = 2Q_{\tilde{J}_t}$.

Short-time existence

Theorem (Cheng-Lee 1990)

Let $\hat{J}$ be any smooth oriented CR structure on (M, ξ). For any integer $m \in \mathbb{N}$ (finite), there exists $\varepsilon > 0$ such that the initial value problem of regularized Cartan flow equation with $J_0 = \hat{J}$ has a solution $J \in C^m$ on time interval $[0, \varepsilon]$.

The proof is based on Folland-Stein space theory.

Theorem (Cheng 2002)

For large enough m (e.g. $m \geq 14$), the C^m-solution of the initial value problem of regularized Cartan flow equation is unique in a small interval $[0, \varepsilon']$ with $\varepsilon' < \varepsilon$.

Short-time existence

Theorem (Cheng 2014)

Suppose J_0 is holomorphic fillable in the convex side with respect to θ with $A_{J_0,\theta} = 0$ and $R_{J_0,\theta} < 0$. Solution to the regularized Cartan flow equation with $K = J_0$ exists on $[0, \varepsilon)$, then $A_{J_t,\theta}$ stays vanishing on $[0, \varepsilon)$, and J_t stays holomorphic fillable in the convex side with respect to θ on $[0, \varepsilon'')$ where $\varepsilon'' \leq \varepsilon$.

Remark

The holomorphically fillability in the convex side with respect to θ means the Reeb direction $\mathcal{R}$ for θ under the action of complex structure of the holomorphic filling points inwards. The parameter ε'' is chosen sufficiently small such that $R_{J_t,\theta} < 0 \ \forall\, t \in [0, \varepsilon'')$.

The End